Building an Interactive Sales Dashboard in Power BI Using DAX and DAX Studio

For this practical, I worked with the Adventure Works dataset in Power BI. This time, the focus was on creating KPIs, writing custom DAX calculations, analysing sales trends and building an interactive dashboard. I also used DAX Studio to check whether my calculations were working correctly.

If you are new to Power BI, DAX is basically the language used to create calculations and measures in Power BI. It helps us go beyond basic charts and get more useful insights from our data.

Part A: Preparing the Data

Before starting with the calculations, I first had to prepare the data. The Adventure Works dataset contained separate sales files for different years, along with tables containing information about products, categories, customers and territories.

I used Power Query to append the yearly sales tables into one table called Sales_Fact. After that, I merged the product, subcategory, category and territory tables to bring all the required information together.

I also checked the data types and made sure the order date was properly connected to the calendar table. This was important because the time-based calculations later depended on this relationship.

Part B: Creating KPIs Using DAX

The next step was to create important business metrics using DAX measures. These measures help summarise the data and make it easier to understand the overall performance.

I created KPIs such as Total Sales, Total Cost, Total Profit, Total Orders, Total Customers, Average Sales, Average Order Value and Profit Margin Percentage.

For example, I used SUMX to calculate total sales by multiplying the order quantity by the product price for each row. I calculated total cost in a similar way and then found total profit by subtracting the total cost from total sales.

I also used DISTINCTCOUNT to count unique orders and customers. Finally, I displayed the KPIs using Card visuals so that the important numbers could be viewed together on one page.

Part C: Exploring More DAX Functions

This part was more interesting because I got to work with different DAX functions and understand how they can be used for analysis.

I created 21 additional measures using functions such as SUM, AVERAGE, COUNT, DISTINCTCOUNT, CALCULATE, FILTER, DIVIDE, MAXX, MINX and RANKX.

Some of the calculations included total quantity sold, average product price, profit per customer, sales per customer and cost per unit. I also calculated sales and profit for orders where the quantity was greater than two.

Using MAXX and MINX, I found the maximum and minimum sales values for individual sales records. With RANKX, I ranked products based on their total sales and quantity sold.

This helped me understand how DAX can be used for more than just basic calculations. It can also filter data, compare values and rank products based on their performance.

Part D: Analysing Sales Over Time

After creating the measures, I moved on to time-based analysis. The aim was to compare monthly sales and understand how sales changed from one month to another.

I created four measures called Monthly Sales, Previous Period Sales, Sales Variance and Sales Growth Percentage.

I used the DATEADD function to calculate sales for the previous month. Sales Variance showed the difference between the current month's sales and the previous month's sales. Sales Growth Percentage helped me understand the relative change in sales.

I displayed these values using Card visuals and a table containing the monthly figures. This made it easier to compare sales across different periods and identify changes in performance.

Part E: Using DAX Studio

DAX Studio was another important part of this practical. It allowed me to execute DAX queries separately and check the results against the values displayed in Power BI.

I connected DAX Studio to my open Power BI model and ran queries to validate the KPI measures. I also queried the time-based calculations, retrieved the top 10 products by sales and compared sales and profit across product categories and regions.

This was useful because it gave me another way to check my calculations instead of relying only on the report visuals. I also got a better understanding of how DAX queries can retrieve and summarise data.

Part F: Building the Interactive Dashboard

The final task was to bring everything together into an interactive dashboard.

I created KPI cards to display Total Sales, Total Profit, Total Orders and Total Customers. I also added five visuals to analyse the data from different perspectives.

These included a column chart showing total sales by region, a map showing customers by country, a donut chart showing profit by product category, a line chart showing monthly sales and a bar chart displaying the top 10 products by sales.

I added slicers for year, product category and region. These allow users to filter the dashboard and view the results for specific selections.

The best part was seeing how the charts and KPI cards responded to the slicers. Instead of looking at separate tables and calculations, I could explore the sales data through one dashboard.

What I Took Away From This Practical

This practical helped me understand how Power BI, DAX and DAX Studio work together. I learned how to create useful business metrics, apply different DAX functions, analyse monthly sales and validate calculations using DAX Studio.

I also realised that building a dashboard involves more than just creating charts. Preparing the data properly and writing the correct measures are equally important.

Overall, this practical gave me a better understanding of how Power BI can be used to turn raw sales data into an interactive dashboard that makes business analysis easier.